

Vendor Performance & Inventory Analysis

Data-Driven Insights for Vendor Selection & Pricing Optimization

Retail / Beverage Distribution Inventory Dataset
SQL (SQLite) • Python (Pandas, Seaborn, SciPy) • Statistical Testing

Repository: github.com/manciii/vendor_analysis

Executive Summary

This analysis examines vendor and product performance for a beverage distribution business using purchase, sales, and inventory data consolidated from five source tables into a single vendor-sales summary. The objective was to identify which vendors and brands drive profitability, where capital is inefficiently tied up in unsold inventory, and whether purchasing strategy (bulk buying, vendor concentration) meaningfully affects cost and margin. The findings support two core business questions: which vendors to prioritize for continued partnership, and which products need pricing or promotional intervention.

\$2.71M Total Unsold Capital	65.7% Top 10 Vendor Purchase Share	DIAGEO (\$68M) Top Vendor Sales	8,565 Records Analyzed (cleaned)
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Key Findings

- Vendor concentration is high: the top 10 vendors account for 65.7% of total purchase dollars, leaving the business exposed if any single relationship is disrupted.
- A statistically significant gap exists between high- and low-volume vendors' profit margins ($p < 0.0001$) — low-volume vendors run notably higher margins (~41.6%) than top-volume vendors (~31.2%), suggesting the highest-selling vendors are being priced too competitively.
- Bulk purchasing meaningfully reduces unit cost: large orders cost \$10.78/unit versus \$39.06/unit for small orders, a ~72% reduction.
- \$2.71M in capital is locked in unsold inventory, concentrated in a handful of vendors (Diageo, Jim Beam, Pernod Ricard, William Grant & Sons).
- A specific set of brands show low sales volume but high profit margins — strong candidates for promotional investment rather than discounting.

1. Data Foundation & Preparation

The source SQLite database (`inventory.db`) contains several raw operational tables: `purchases`, `purchase_prices`, `vendor_invoice`, `sales`, plus beginning/ending inventory snapshots. The exploratory phase (*Exploratory_Data_Analysis.ipynb*) mapped how these tables relate to one another before consolidating them into a single, analysis-ready table.

Table Relationships

- **purchases** — transaction-level purchase records: date, vendor, brand, quantity, and dollar amount.
- **purchase_prices** — one row per vendor/brand combination with list price and volume; used to attach actual pricing.
- **vendor_invoice** — PO-level aggregation of purchases with freight cost, unique by vendor and PO number.
- **sales** — transaction-level sales records: quantity, dollars, price, and excise tax by vendor and brand.
- Beginning/ending inventory tables were excluded from the vendor-behavior analysis as out of scope.

Building the Vendor Sales Summary

A single consolidated table, `vendor_sales_summary`, was built by joining freight, purchase, and sales summaries on vendor and brand. This pre-aggregation avoids repeating expensive joins on large tables for every future query and forms the base table for both dashboarding and this analysis. Derived metrics were added on top of the raw aggregates:

- **GrossProfit** = $\text{TotalSalesDollars} - \text{TotalPurchaseDollars}$
- **ProfitMargin** = $\text{GrossProfit} / \text{TotalSalesDollars} \times 100$
- **StockTurnover** = $\text{TotalSalesQuantity} / \text{TotalPurchaseQuantity}$
- **SalesToPurchaseRatio** = $\text{TotalSalesDollars} / \text{TotalPurchaseDollars}$

Data Cleaning

Cleaning addressed three issues found during profiling: the `Volume` column was stored as text and converted to numeric; missing values (mainly products purchased but never sold) were filled with zero; and leading/trailing whitespace was stripped from vendor name and description fields to prevent duplicate groupings. The cleaned table was written back to the database for reuse.

2. Exploratory Data Analysis

Before drawing business conclusions, the consolidated table was profiled for distribution shape, outliers, and data-quality issues that could distort downstream analysis.

Distribution & Outlier Screening

Histograms and boxplots were generated across all numerical fields. Several data-quality flags emerged: gross profit had a minimum of $-\$52,002.78$ (loss-making transactions), profit margin included cases of zero or negative revenue, and a subset of products were purchased but recorded zero sales — consistent with slow-moving or obsolete stock. Purchase and actual prices also showed extreme right-skew, with maximum values ($\$5,681.81$ and $\$7,499.99$) far above the means ($\24.39 and $\$35.64$), indicating a small number of premium or specialty products.

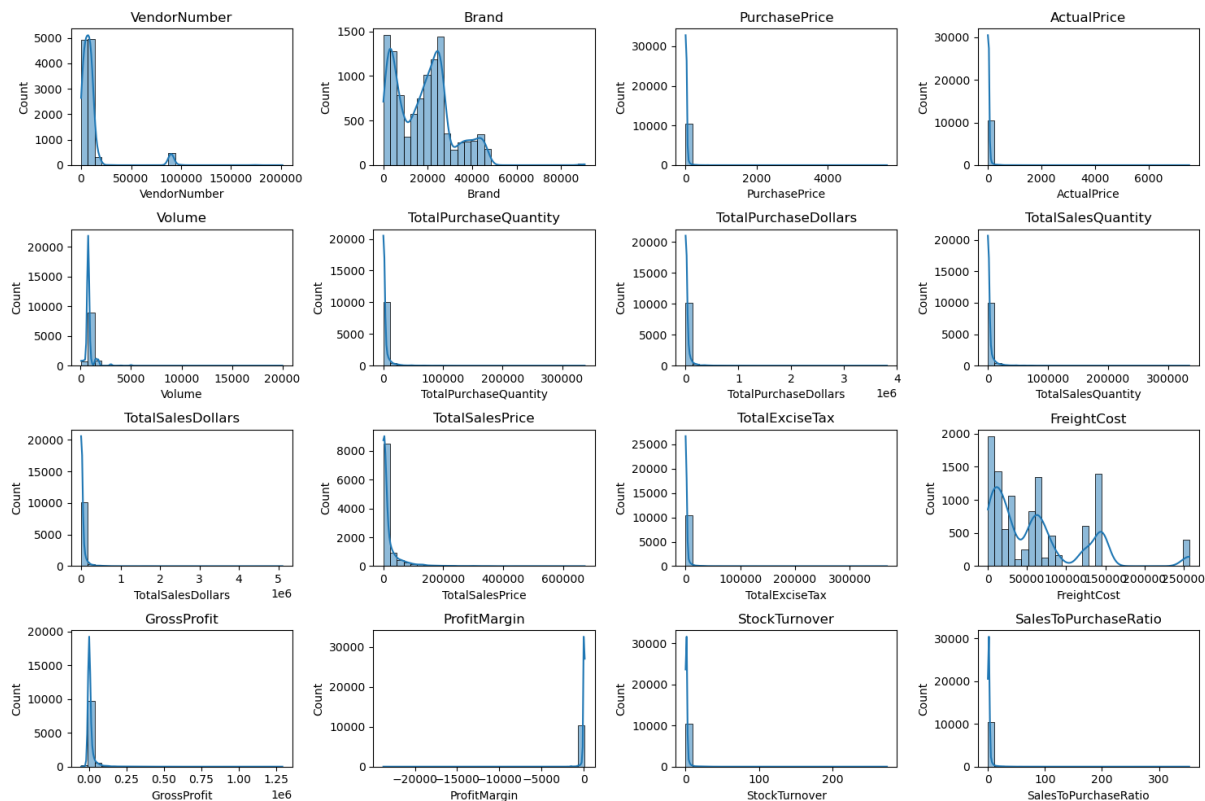


Figure 1. Distribution of numerical fields prior to cleaning — note the long right tails on price and dollar fields.

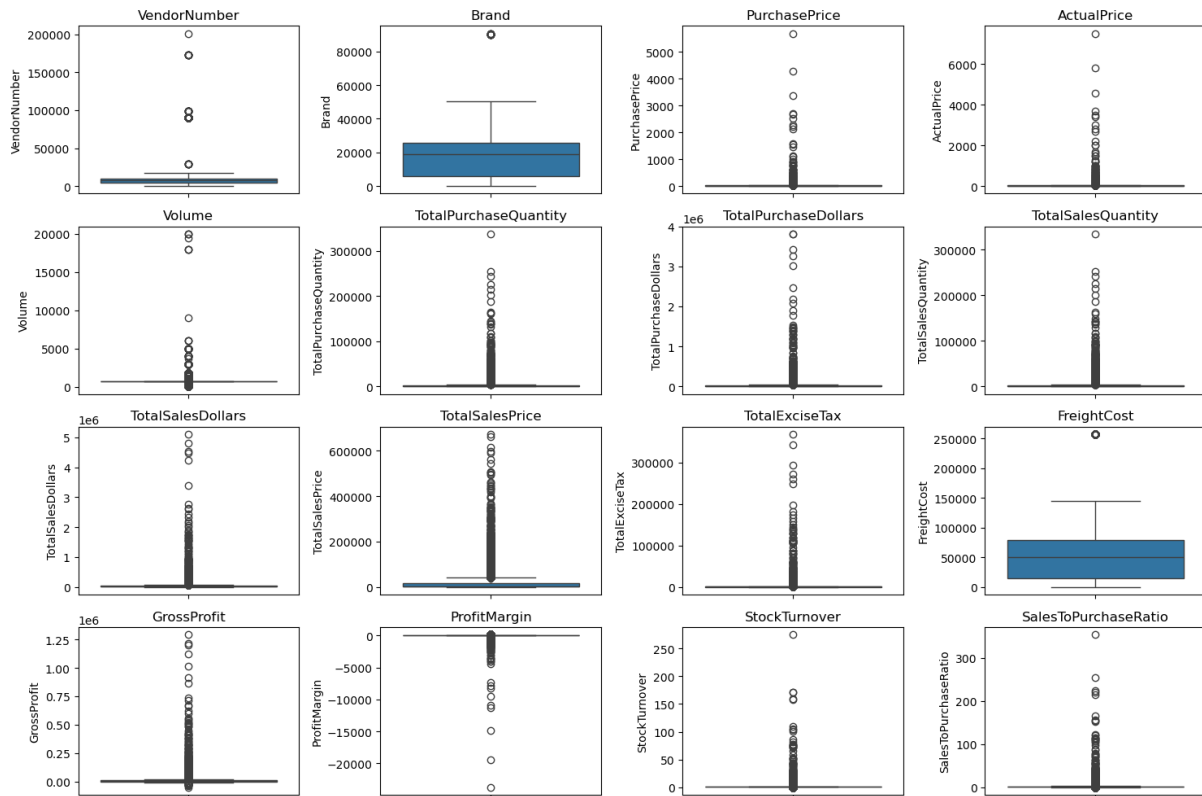


Figure 2. Boxplots highlighting outliers across purchase price, sales price, and profit fields.

To keep the profitability analysis meaningful, records with non-positive gross profit, non-positive profit margin, or zero sales quantity were filtered out, reducing the working dataset to 8,565 rows. Re-running the distribution check on the filtered data confirmed the extreme negative values were removed while the underlying shape (right-skewed pricing, concentrated volume) remained intact and representative of normal business activity.

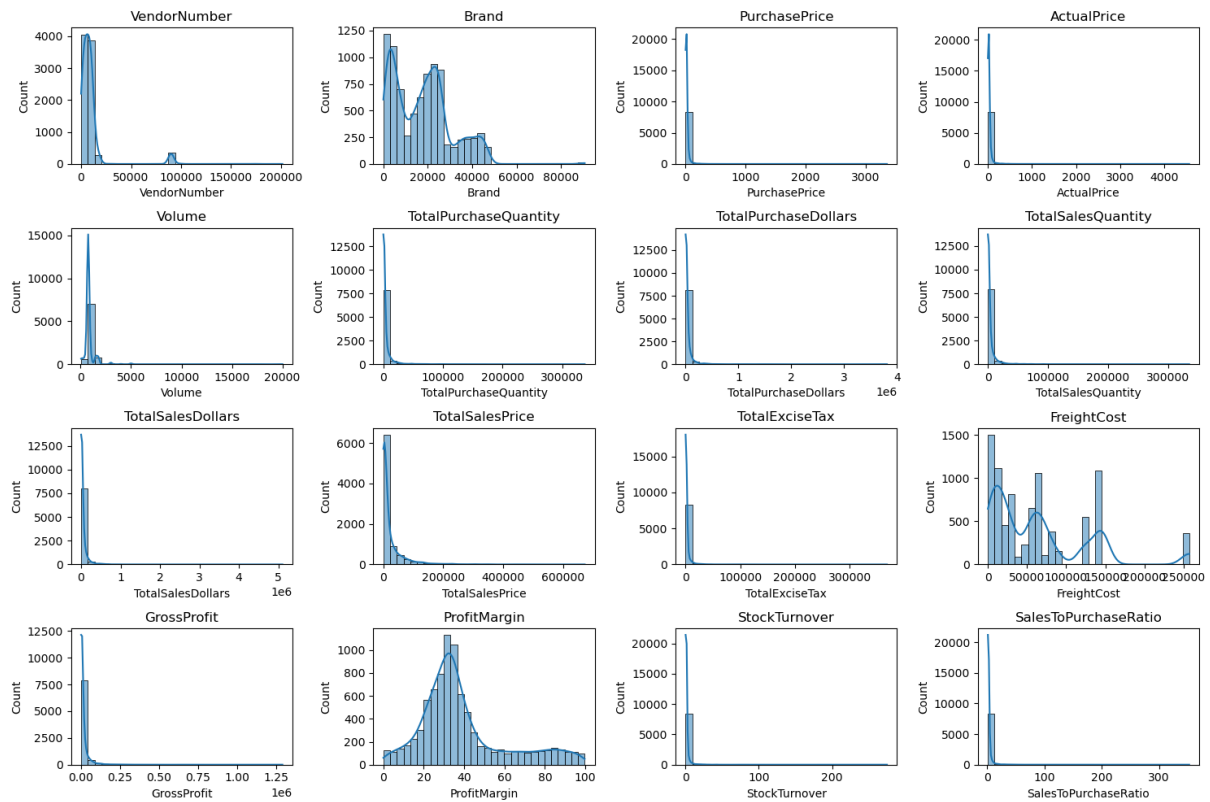


Figure 3. Distributions after removing loss-making and zero-sales records.

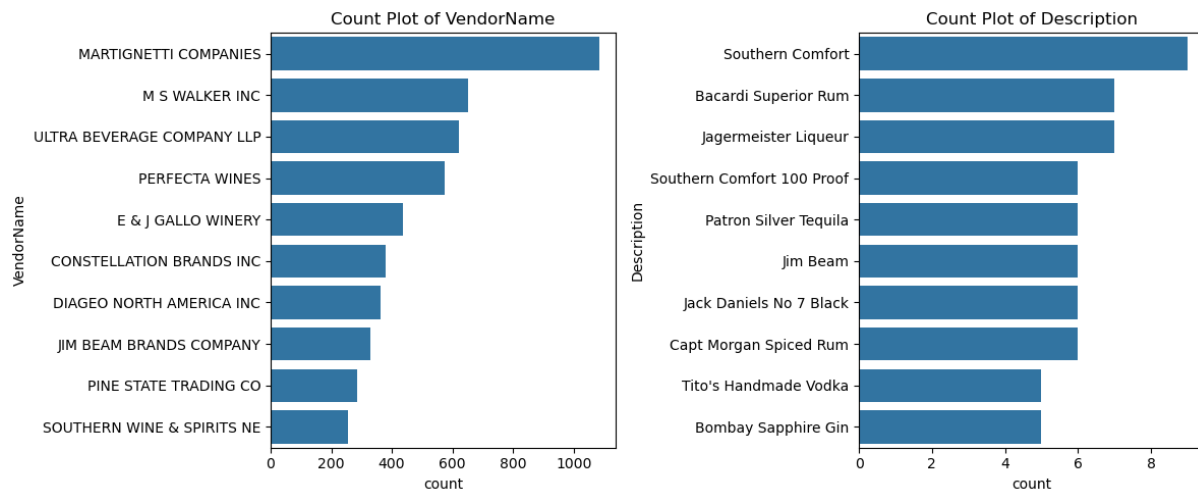


Figure 4. Top 10 vendors and product descriptions by record count.

Correlation Analysis

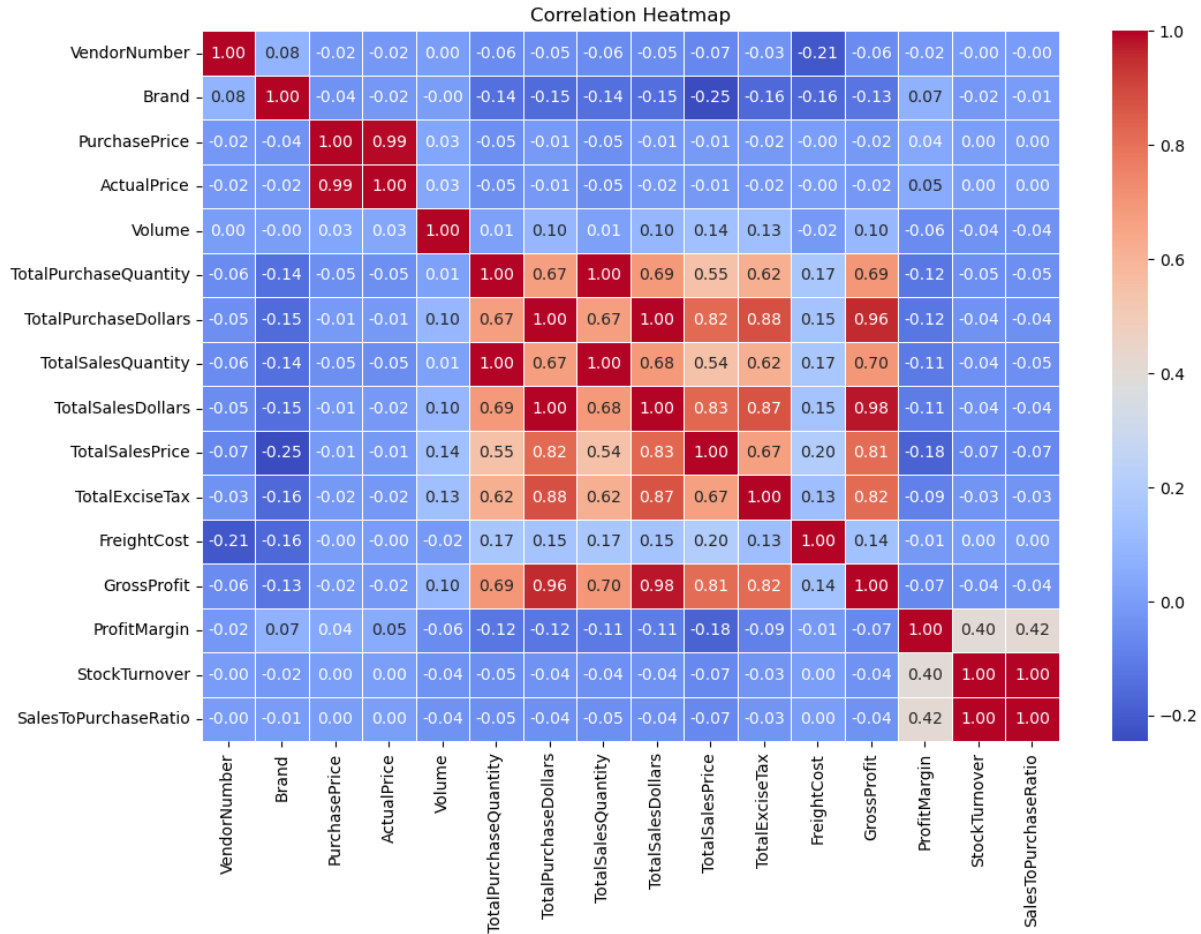


Figure 5. Correlation heatmap across all numerical fields.

- PurchasePrice has almost no correlation with TotalSalesDollars (−0.01) or GrossProfit (−0.02), meaning price alone does not drive revenue or profit outcomes.
- TotalPurchaseQuantity and TotalSalesQuantity move together almost perfectly (0.999), confirming efficient inventory flow-through overall.
- ProfitMargin is negatively correlated with TotalSalesPrice (−0.18), suggesting margins compress as sales prices rise — likely a sign of competitive pricing pressure on higher-priced items.
- StockTurnover shows weak negative correlation with both GrossProfit (−0.04) and ProfitMargin (−0.06): moving inventory faster does not automatically translate into higher profitability.

3. Vendor & Brand Sales Performance

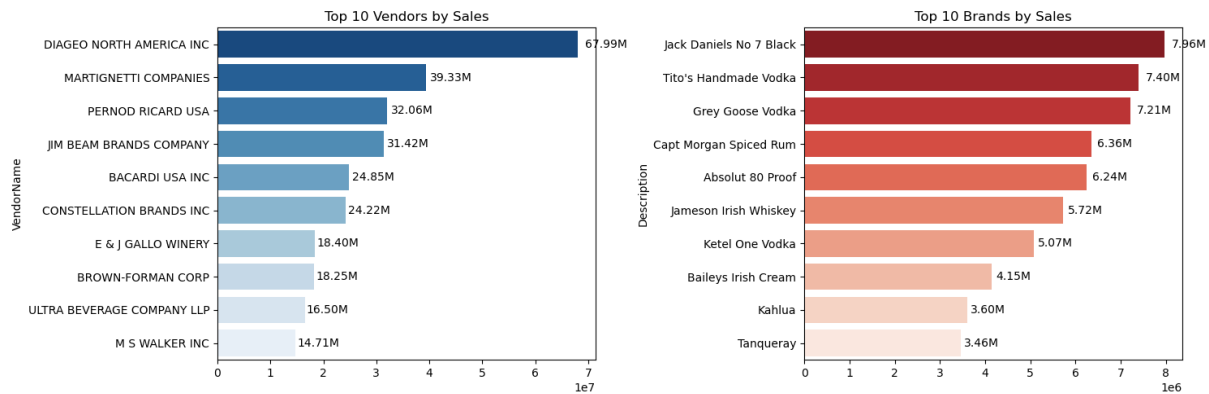


Figure 6. Top 10 vendors and top 10 brands ranked by total sales dollars.

Diageo North America leads all vendors with \$67.99M in sales, nearly double the second-place Martignetti Companies (\$39.33M). At the brand level, Jack Daniels No. 7 Black (\$7.96M) and Tito's Handmade Vodka (\$7.40M) are the top revenue generators, with a fairly even drop-off across the rest of the top 10 — no single brand dominates the way Diageo dominates at the vendor level.

Brands for Promotional or Pricing Adjustment

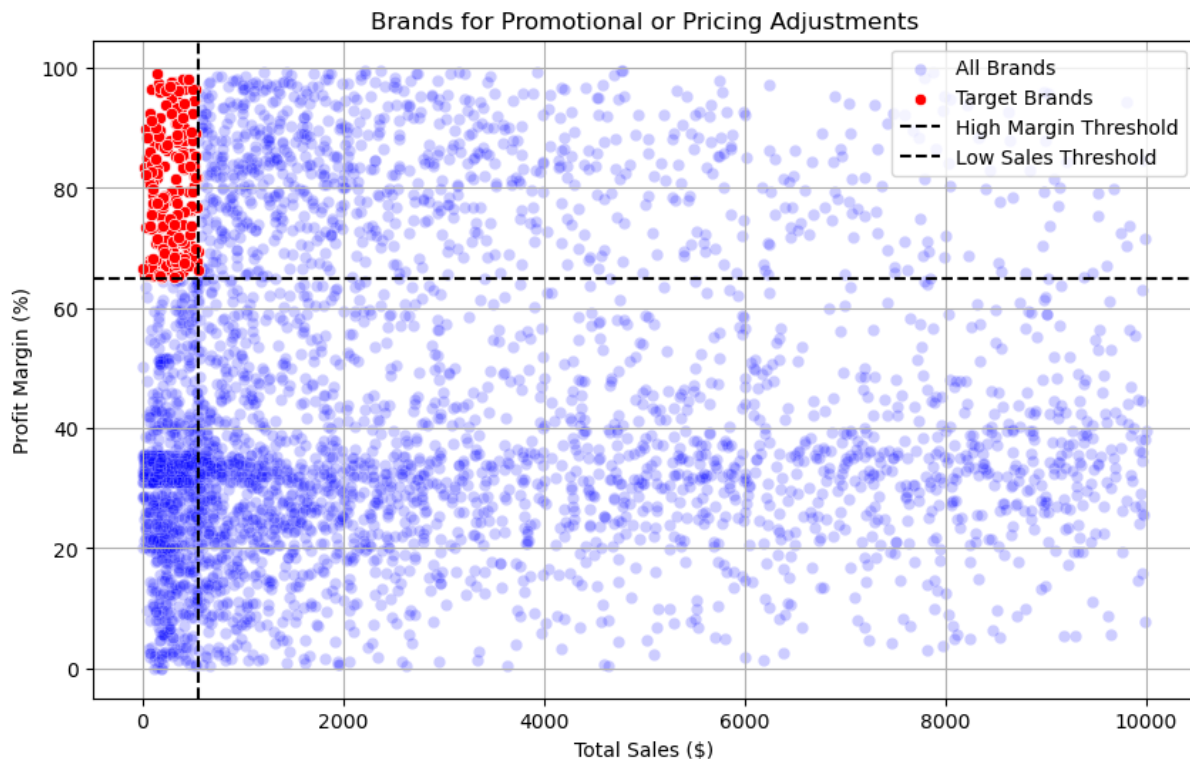


Figure 7. Brands with low total sales but high profit margin (red), flagged against the full population (blue).

Using the bottom 15% of brands by total sales dollars and the top 15% by profit margin as thresholds, a distinct cluster of brands emerges that sell in low volume yet retain strong margins — for example Pepperjack Barossa Red, Flint & Steel Sauvignon Blanc, and Croft Tawny Porto. These are strong candidates for

promotional investment (to grow volume while margin is still healthy) rather than markdown, since cutting price further would erode the one advantage they currently have.

4. Vendor Concentration & Procurement Risk

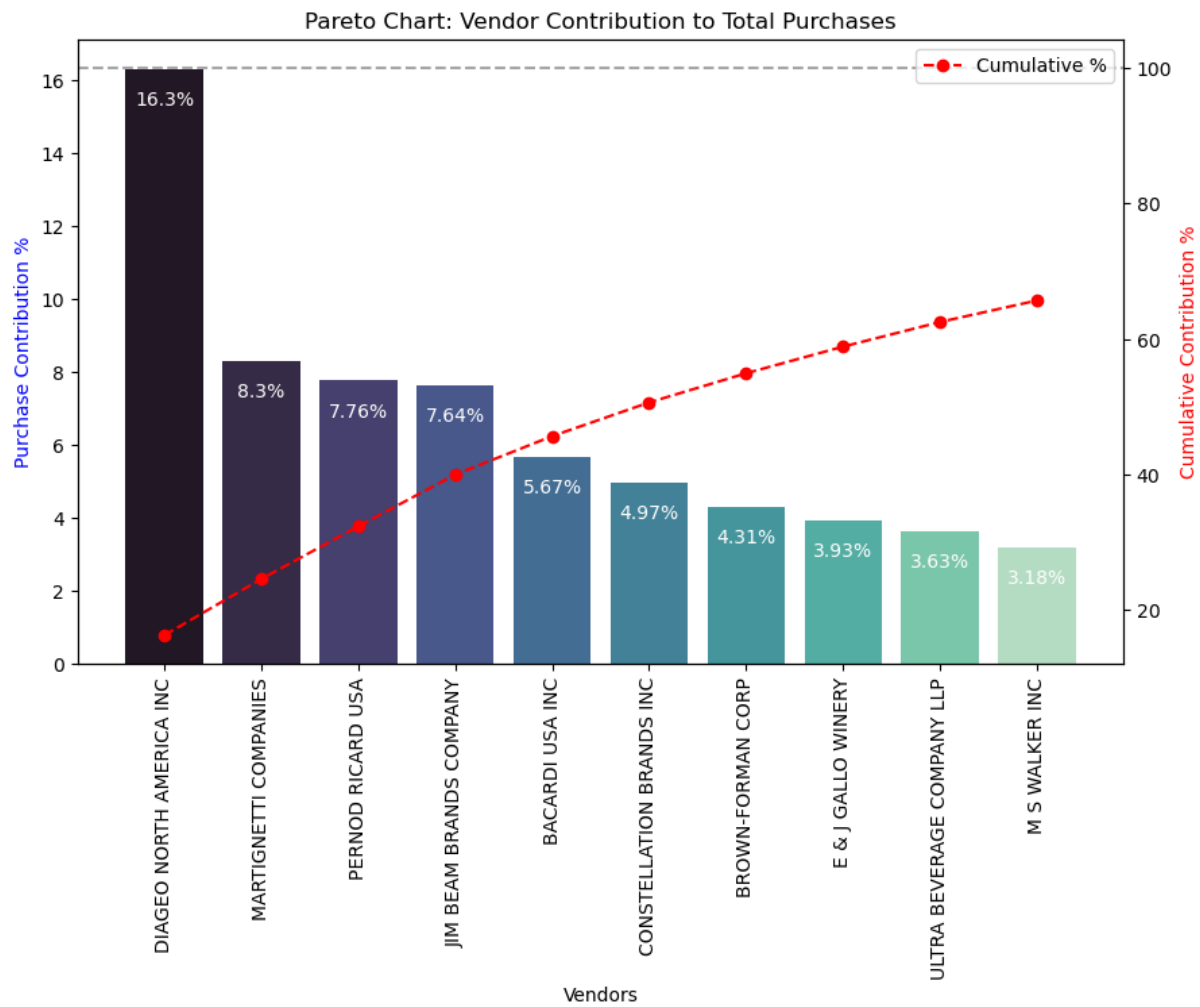


Figure 8. Purchase contribution (%) of the top 10 vendors with cumulative contribution line.

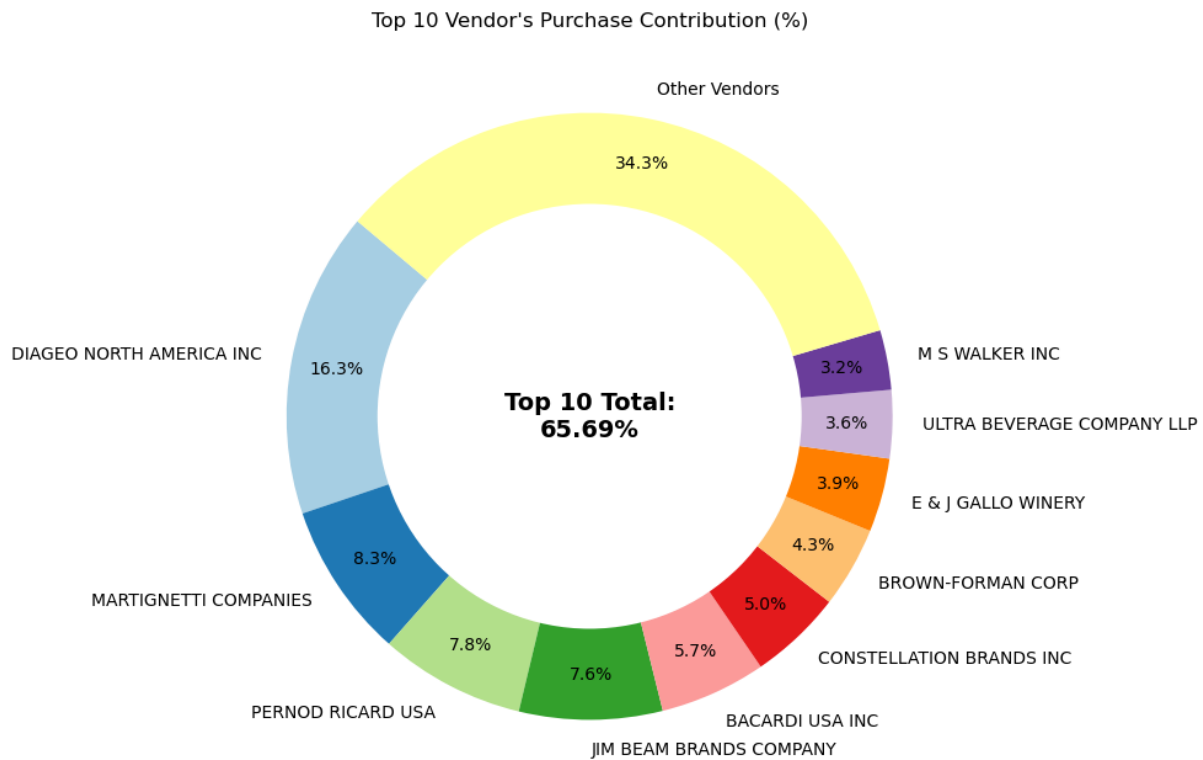


Figure 9. Share of total procurement dollars concentrated in the top 10 vendors vs. all others.

The top 10 vendors account for 65.7% of total procurement spend, led by Diageo North America alone at 16.3%. The remaining 34.3% is spread across the rest of the vendor base. This level of concentration is a double-edged sword: it likely brings volume-based pricing leverage, but it also creates meaningful supply risk — a disruption at any one of the top few vendors would have an outsized effect on procurement. Where dependency on a small vendor group is this high, it is worth evaluating whether a broader, more diversified supplier base would reduce risk without materially increasing cost.

5. Bulk Purchasing & Unit Cost

Purchase orders were grouped into three equal-sized tiers (Small, Medium, Large) by total purchase quantity to test whether ordering in larger volumes reduces unit price.

Order Size	Avg. Unit Purchase Price
Small	\$39.06
Medium	\$15.49
Large	\$10.78

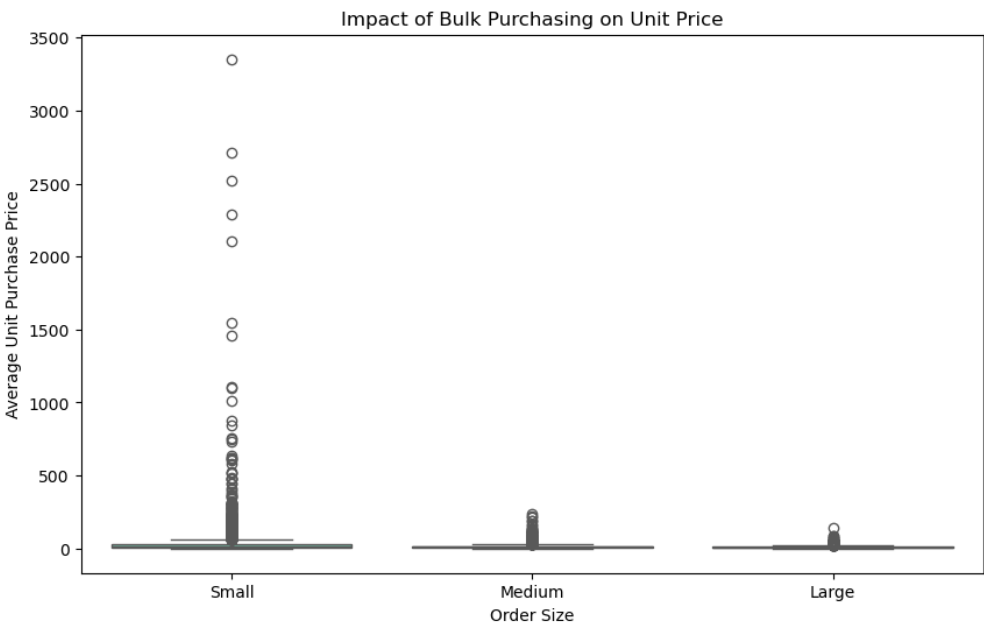


Figure 10. Unit purchase price by order size tier.

The relationship is clear and substantial: large orders cost \$10.78 per unit versus \$39.06 for small orders — a roughly 72% reduction. This confirms that bulk-purchasing strategies are effective and gives a concrete basis for consolidating smaller, frequent orders into larger, less frequent ones wherever storage and cash-flow constraints allow.

6. Inventory Efficiency & Locked Capital

Stock turnover (sales quantity / purchase quantity) was used to flag vendors whose products move slowly through inventory. Vendors with turnover below 1.0 are, on average, purchasing faster than they are selling.

Lowest Stock Turnover Vendors

Vendor	Avg. Stock Turnover
Alisa Carr Beverages	0.62
Highland Wine Merchants LLC	0.71
Park Street Imports LLC	0.75
Circa Wines	0.76
Dunn Wine Brokers	0.77
Centeur Imports LLC	0.77

Capital Locked in Unsold Inventory

Unsold inventory value was estimated as (purchase quantity – sales quantity) × purchase price. Across the business, an estimated **\$2.71M** of capital is currently tied up in unsold stock. This is concentrated among a small number of large vendors:

Vendor	Unsold Inventory Value
Diageo North America Inc	\$722.21K
Jim Beam Brands Company	\$554.67K
Pernod Ricard USA	\$470.63K
William Grant & Sons Inc	\$401.96K
E & J Gallo Winery	\$228.28K
Sazerac Co Inc	\$198.44K

Slow-moving inventory carries real cost — warehouse space, insurance, and depreciation — even before accounting for the opportunity cost of the capital itself. Because these figures are concentrated in otherwise high-volume, high-revenue vendors, the issue is less about vendor selection and more about SKU-level demand forecasting and order-quantity discipline within those relationships.

7. Statistical Analysis: Top vs. Low-Performing Vendors

Vendors were split into top-performing (top 25% by total sales dollars) and low-performing (bottom 25%) groups to test whether profit margins genuinely differ between them, rather than differing by chance.

95% Confidence Intervals for Profit Margin

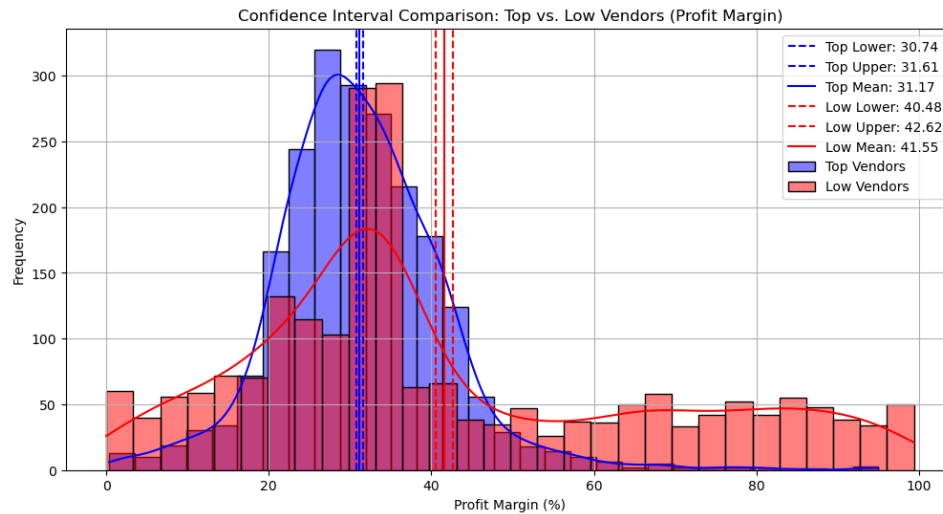


Figure 11. Distribution and 95% confidence intervals for profit margin, top vs. low-performing vendors.

Group	Mean Profit Margin	95% Confidence Interval
Top-performing vendors	31.17%	(30.74%, 31.61%)
Low-performing vendors	41.55%	(40.48%, 42.62%)

Hypothesis Test

H₀ (Null): There is no significant difference in mean profit margin between top-performing and low-performing vendors.

H₁ (Alternative): The mean profit margins differ significantly.

A Welch's two-sample t-test was run comparing the two groups' profit margins.

Statistic	Value
T-statistic	-17.64
P-value	< 0.0001
Conclusion	Reject H0

The result is highly statistically significant: low-performing (by sales volume) vendors maintain meaningfully higher profit margins than top-performing vendors, and this gap is far too large to be explained by random variation. Practically, this points to a pricing trade-off — the vendors driving the most volume are likely doing so partly through more aggressive/competitive pricing, while lower-volume vendors sustain premium pricing at the cost of scale.

8. Recommendations

Pricing & Promotion

- Explore selective price increases or bundling for top-volume vendors, where margin has room to improve without materially hurting the volume they already command.
- Invest promotional spend behind the identified low-sales/high-margin brands (Figure 7) instead of discounting — the goal is to grow their volume while their margin advantage is still intact.

Procurement Strategy

- Consolidate smaller, frequent orders into larger batch purchases where feasible — the unit-cost savings from bulk buying (~72% reduction, large vs. small orders) are substantial.
- Evaluate supplier diversification given that 65.7% of purchase spend sits with just 10 vendors; assess whether qualifying 2–3 additional suppliers in Diageo's or Martignetti's categories would reduce concentration risk.

Inventory Management

- Prioritize demand-forecasting review for the vendors with the largest unsold-inventory balances (Diageo, Jim Beam, Pernod Ricard, William Grant & Sons) — these are high-volume relationships where even small order-quantity corrections free up significant capital.
- Set a stock-turnover threshold (e.g. below 1.0) as a standing flag for slow-moving vendor/brand combinations, and route flagged items to a re-order-quantity review before the next purchase cycle.

Analysis based on Exploratory_Data_Analysis.ipynb and Vendor_Performance_Analysis.ipynb. All figures reflect the cleaned vendor_sales_summary dataset (8,565 records) after removing loss-making and zero-sales transactions.